

Claes Johnson, Professor emeritus of Applied Mathematics
KTH Royal Institute of Technology, Stockholm | claesjohnson@gmail.com

To the Editors, *Studies in History and Philosophy of Science*

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Dear Editors,

I submit for your consideration the article *Quantum Mechanics without Ontology? Configuration Space, the Arena Problem, and the Road Not Taken in 1926*, for publication in *Studies in History and Philosophy of Science*. The paper sits squarely at the intersection of the history and the philosophy of modern physics that the journal exists to serve: it combines a reading of the 1926 turn to configuration space with an argument about the metaphysics of the quantum state.

Its contribution is threefold. *First*, it sharpens the common claim that quantum mechanics “abandons ontology.” Taking the Pusey–Barrett–Rudolph theorem seriously, the paper grants that the quantum state is plausibly ontic, and relocates the difficulty from the *existence* of an ontology to its *arena*: the state that PBR renders real is a field on $3N$ -dimensional configuration space, not on the three-dimensional space of observers and instruments. The measurement problem, collapse, the Heisenberg cut, and the nonlocality puzzles are then presented as symptoms of this *arena problem* rather than of a bare absence of the real. *Second*, it situates the resulting options explicitly within the contemporary debate between wave-function realism (Albert, Ney, Wallace) and primitive-ontology approaches (Allori, Dürr–Goldstein–Zanghì, Maudlin, Bell), and it revisits the historical moment — with reference to Schrödinger’s correspondence with Lorentz and to Mehra and Rechenberg — at which the configuration-space arena was chosen over the objections of Schrödinger and Einstein, arguing that the choice was contingent rather than forced. *Third*, it offers a concrete three-dimensional charge-density formulation as an under-explored member of the primitive-ontology family — an existence proof that the road not taken in 1926 remains open — and draws a practical corollary for quantum computing, whose claimed advantage turns on the physical reality of the exponentially large configuration-space amplitude.

The paper is, I believe, of interest to historians and philosophers of physics alike: it uses a precise recent result (PBR) and a live metaphysical debate (wave-function realism versus primitive ontology) to give the century-old interpretive impasse a clearer diagnosis, and it grounds the philosophical claim in a datable historical episode. The manuscript is original, has not been published elsewhere, and is not under consideration by another journal.

In the interest of full transparency: the manuscript was prepared in cooperation with an AI assistant (Claude, Anthropic), which contributed to drafting and to surveying the secondary literature; the argument, claims, and responsibility for the content are the author’s. I am glad to describe this collaboration further if the journal wishes.

Thank you for your consideration.

Sincerely,
Claes Johnson